Lean Six Sigma Practice Test

1. The purpose of Six Sigma is to:

   A. Fix defects.
   B. Reduce variation.
   C. Perform project management.
   D. Calculate the standard deviation.

2. What is Six Sigma?

   A. A business philosophy focusing on improvement
   B. A project management model
   C. A defects reduction technique that is six times better than average
   D. A measurement system

3. DMAIC stands for:

   A. Design, Measure, Analyze, Improve, Control
   B. Define, Measure, Analyze, Improve, Control
   C. Double Measurement And Instructions Checklist
   D. Definition, Measurements, Analysis, Improvements, and Control

4. DFSS stands for:

   A. Design and Fix using Six Sigma
   B. Develop Fast using Six Sigma
   C. Define Futuristic Service and Solutions
   D. Design For Six Sigma

5. Six Sigma Level can be achieved by having:

   A. Less than 3.4 defects per million opportunities.
   B. Six times better quality than competitors.
   C. Six levels of testing and validation before release.
   D. Six Nines Quality (99.9999%).

6. DMADV stands for:

   A. Design, Measure, Analyze, Develop, Verify
   B. Develop Measurement and Analysis Department for Verification
   C. Define, Measure, Analyze, Design, Verify
   D. Design, Measure, Analyze, Define, Validate

7. The purpose of the Toll Gate review is to:

   A. Confirm if the earlier phase was completed successfully.
   B. Get a formal sign-off from the stakeholders.
   C. Review progress and kick off the next phase.
   D. All of the above

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8. What is the primary role of Master Black Belt?

A. Provide a budget for the projects
B. Coach, mentor, and train Black Belts
C. Be a master in Six Sigma implementation
D. Review projects

9. Which role performs the following responsibilities: "Evangelize Six Sigma, provide resources and resolve any cross-functional issues"?

A. Sponsor
B. Senior Executive
C. Master Black Belt
D. Champion

10. A Black Belt is expected to deliver what types of results from Six Sigma projects?

A. At least 200,000 USD as the annual benefit
B. Have everybody in the organization trained on Six Sigma
C. Large projects
D. Defects reduction projects

11. The purpose of capturing Voice of Customer is to:

A. Ensure there is no confusion between the contract and delivery.
B. Gather customer feedback.
C. Have a sampling exercise.
D. Capture stated or unstated customer needs or requirements.

12. In which role is a full-time professional to lead large Six Sigma projects?

A. Six Sigma Champion
B. Green Belt
C. Master Black Belt
D. Black Belt

13. What is the role of Black Belts in dealing with organizational roadblocks?

A. Escalating issues to Senior Management
B. Communicating the benefits of Six Sigma, changing efforts and building confidence in the team
C. Asking for help from Master Black Belts to handle a situation
D. Providing Six Sigma training to everybody in the organization

14. Why do Black Belts and Master Black Belts conduct closed-looped assessments of Six Sigma implementation?

A. To ensure all aspects of Six Sigma have been implemented
B. To assure the privacy of the people involved in Six Sigma projects
C. To ensure competitors do not get the information on the improvements achieved
D. To minimize influencing people with the Six Sigma project measurements and avoid any manipulation

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15. What tool did Kaoru Ishikawa develop?

A. SWAT Analysis Tool
B. CTQ Diagram
C. Cause and Effect Diagram
D. FMEA Tool

16. What is the best mechanism to collect VOC data?

A. Complaints Database
B. Customer Support History
C. Industry Reports
D. Surveys and Questionnaire

17. Which of the following Six Sigma tools can be used by a team to prioritize a list of options to improve?

A. Brainstorming
B. Nominal Group Technique
C. Cause and Effect Diagram
D. 5 Whys

18. What does SWOT stand for?

A. Strengths, Weaknesses, Opportunities, Threats
B. Six Work Optimization Techniques
C. Seven Workplace Opportunities for Transformation
D. Strengths, Weaknesses, Opportunities, Techniques

19. The primary factor in the successful implementation of Six Sigma is to have:

A. Training for everybody in the organization
B. Support from everybody in the organization
C. Experienced Six Sigma practitioners
D. Executive commitment and support

20. What is Continual Improvement Process?

A. The principle of never stopping any improvements
B. Improvements being made while people and machines continue to work
C. A steady effort made to improve existing processes, products, or services
D. The non-stop cyclic process of improvements where one improvement leads to another

21. What type of chart one would typically use for analyzing positional, cyclical, and temporal variations?

A. Multi-Vari Charts
B. SPC Charts
C. Cause and Effect Diagram
D. Run Charts

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22. If you need to collect data to analyze the symptoms of a problem, what tool will be most appropriate?

A. Flow Chart
B. FMEA Tool
C. Force Field Analysis
D. Check Sheet

23. The purpose of having functional requirements is:

A. To describe a single, measurable performance.
B. To provide details on how a product or service needs to operate.
C. To provide specification limits.
D. To build VoC.

24. Increasing performance in a Six Sigma project from 3 sigma to 4 sigma would reduce defects per million by a factor of:

A. 2
B. 8
C. 10
D. 16

25. In a Cause and Effect diagram, some of the major cause categories are mother nature, management, measurement systems, and the following four standard causes:

A. Marketing, methods, material, machines
B. Man, material, methods, machines
C. Man, manufacturing, methods, material
D. Man, material, medicines, machines

26. To monitor the number of defects on different sample sizes, which of the following control charts is most appropriate to use?

A. np chart
B. c chart
C. u chart
D. p chart

27. What are the 5 elements of Kaizen?

A. Teamwork, Personal Discipline, Improve Morale, Quality Circles, Suggestions for Improvement
B. Define, Measure, Analyze, Improve and Control
C. Define, Measure, Analyze, Design and Validate
D. Identify Value, Map the Value Stream, Create Flow, Establish Pull, Seek Perfection

28. What are the 5 Lean principles?

A. Identify Value, Map the Value Stream, Create Flow, Establish Pull, Seek Perfection
B. Teamwork, Personal Discipline, Improve Morale, Quality Circles, Suggestions for Improvement
C. Define, Measure, Analyze, Improve, and Control
D. Reduce Waste, Organize, Just-In-Time, optimize, Improve

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29. Measurement accuracy is attained when:

A. Different people get the same result when measuring the same item or characteristic.
B. The same person taking multiple measurements on the same item or characteristic gets the same results every time.
C. The measured value has little deviation from the actual value.
D. The resolution of the measurement instrument can give at least 5 distinct values in the range being measured.

30. In a Normal distribution, approximately 68% of the data will occur within:

A. +/- 2 standard deviation
B. +/- 1 standard deviation
C. +/- 3 standard deviation
D. Outside the bell-shaped curve

31. In FMEA exercise, the Risk Priority Number (RPN) is calculated as:

A. Severity x occurrence x risk
B. Severity x risk x detection
C. Occurrence x detection x control
D. Severity x occurrence x detection

32. Which of the following activities is a value-added activity?

A. Testing and Inspection
B. Setup
C. Process
D. Storage

33. What does Lean help you with?

A. Reducing weight
B. Eliminating waste and reducing non-value adding activities
C. Achieve higher quality
D. Improving employee morale

34. We want to test the performance of people before and after training. Which of the following tests can be used if the training improved workers performance?

A. 2-sample z test
B. Null hypothesis
C. Paired t-test

35. Which of the following is a chart that depicts the frequencies of measurement data?

A. Histogram
B. Run Charts
C. Control charts
D. Pie Chart

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36. The purpose of using control charts is to:

A. Detect the root causes of defects.
B. Validate Measurements System.
C. Control the output of a process.
D. Evaluate process performance over time.

37. Design for Test is a design technique that offers a benefit of:

A. Customer Satisfaction Tests
B. Defect Diagnosis
C. Manufacturability
D. Smaller Process Variation

38. What does DFMEA stand for?

A. Design Failure Mode and Effect Analysis
B. Defect Failure Mode and Effect Analysis
C. Derive Failure Mode and Estimation Analysis
D. Design Failure Mode and Estimation Analysis

39. Which chart is used to prioritize the highest overall score received as part of the CE matrix?

A. Glaphira Charts
B. Standard Bar Charts
C. Putoro Charts
D. Pareto Charts

40. What of the following is a basic pre-condition for Six Sigma to succeed?

A. Qualified resources
B. Well-defined process
C. Trusted suppliers
D. Reliable funds

41. What does SIPOC stand for?

A. Stakeholders, Inputs, Process, Outputs, and Clients
B. Stakeholders, Inputs, Process, Outputs, and Customers
C. Suppliers, Inputs, Process, Outputs, and Customers
D. Suppliers, Inputs, Purchase, Outputs, and Customers

42. Which question needs to be asked before preparing the SIPOC?

A. Who is the stakeholder for the process?
B. What are the inputs?
C. What resources does the process use?
D. All of the above

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43. What are the various types of measurement scales?

A. Nominal, Ordinal, Interval, and Ratio
B. Nominal, Ordinal, Inverse, and Proportional
C. Direct, Indirect, Inverse, and Proportional
D. Regular, Variance, Inverse, and Proportional

44. What does nominal measurement define or classify?

A. Information about inputs from stakeholders
B. Set of mandatory processes
C. Presence or absence of a certain attribute
D. Minimal length of the project

45. Which measurement scale is a comparative scale of management?

A. Ratio
B. Interval
C. Variance
D. Ordinal

46. What does Ratio Scale define?

A. The ratio of supply vs. demand
B. The ratio of supporters vs. resistors
C. All the true zero value points.
D. Only the positive measurements.

47. Data can be categorized into which of these divisions?

A. Continuous or Variable
B. Discrete
C. Attribute
D. All of the above

48. Which is the most popular and most often used measure of Central Tendency in Six Sigma applications?

A. Mode
B. Mean
C. Median
D. Variance

49. Which measure of Central Tendency in Six Sigma applications should be used when the data has outliners?

A. Mode
B. Mean
C. Median
D. Variance

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50. What are the measures of dispersion?

A. Mean, Mode, and Median
B. Range , Variation, and Standard Deviation
C. Mode, Medium, and Range
D. Median , Range, and Variation

51. By which tool can measurement system accuracy be verified?

A. MSA or GAGE RR
B. CSA or GAGE RR
C. MSA or RAGE RR
D. CSA or RAGE RR

52. What are the different categories in which measurement system errors can be classified?

A. Accuracy and Precision
B. Clarity, Accuracy, and Precision
C. Resolution, Accuracy, and Precision
D. Clarity and Resolution

53. What are the 3 components associated with measurement systems accuracy?

A. Repeatability, Bias, and Linearity
B. Stability, Bias, and Linearity
C. Repeatability, Reproducibility, and Linearity
D. Stability, Bias, and Reproducibility

54. What are the 2 components associated with measurement systems precision?

A. Stability and Bias
B. Bias and Linearity
C. Repeatability and Reproducibility
D. Stability and Reproducibility

55. Which one of these is not a property of a good measurement system?

A. Stability
B. Repeatability
C. Bias
D. Linearity

56. A standard with a known value of 25.4 mm is checked 10 times by one mechanical inspector using a dial caliper with a resolution of 0.025 mm. The readings obtained are: 25.425, 25.425, 25.400, 25.400, 25.375, 25.400, 25.425, 25.400, 25.425, 25.375. The tolerance value of this gage is ± 25 mm. Calculate the Bias value and % Bias.

A. A. 0.05, 1%
B. B. 0.01, 0,2%
C. C. 0.1, 10%
D. D. 0.5, 50%

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57. Which chart indicates Part Variation?

A. Puerto Chart
B. Glaphira Chart
C. Pareto Chart
D. Xbar/Averages Chart

58. What formula is used to measure Total Process Variation(?t) in terms of Gage(?m) and Part(?p) Variation?

A. (σm + σp)/2
B. √(σm2+ σp2)
C. (σm + σp)
D. D. ((σm + σp) * 2)1/2

59. What is Gage Variation(?m) in terms of Repeatability(?r) and Reproducibility(?o) variation?

A. (σr + σo)/2
B. (σr + σo)
C. (σr squared+ σo squared)1/2
D. ((σr + σo)/2))1/2

60. What is the percentage threshold for GRR above which the gage needs to be replaced?

A. A. 0.25
B. B. 0.3
C. C. 0.35
D. D. 0.4

61. While selecting a sample, what rules should govern your choice?

A. Any sample size, if randomly selected, can be suitable for analysis. Since it is sampling, it is not expected to have a specific number of samples.
B. 10% of the total
C. One sample every hour
D. Determine sample size using formula and base it on the tolerance and confidence interval expected and standard deviation or the current error rate of the population

62. Who is the process expert that does the rating of the samples, i.e., as okay, defective, etc.?

A. Master Champion
B. Master Appraiser
C. Process Champion
D. Process Appraiser

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63. What does SPC stand for?

A. Statistical Process Control
B. Standard Process Control
C. Statistical Process Chart
D. Standard Process Chart

64. What does SEM stand for?

A. Standard Error of Mean
B. Standard Error of Mode
C. Standard Estimation of Mode
D. Standard Estimation of Mean

65. Which of the following is not a capability index for continuous data?

A. Process Capability
B. Process Capability Index
C. Process Performance
D. Process Performance Index

66. What is the formula for calculating Process Capability?

A. (USL-LSL)/(6 * σ)
B. (USL+LSL)/(6 * σ)
C. (USL+LSL)/(σ)
D. (USL+LSL)/(2*σ)

67. How do you derive the Process Capability Index (Cpk) once you have the Cpl and Cpu values?

A. min (Cpl, Cpu)
B. max (Cpl, Cpu)
C. avg (Cpl, Cpu)
D. stddev (Cpl, Cpu)

68. When is the Mean not centered?

A. Cpk > Cp
B. Cpk = Cp
C. Cpk < Cp
D. None of the above

69. Which one of these activities should ideally come first in the implementation of a Measure phase?

A. Checking Normality
B. Checking Precision
C. Checking Repeatability
D. Checking Resolution

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70. In a typical Measurement System Analysis activity, the GRR is 42% and the contribution is split equally among Gage Variation and Part Variation. What would you do?

A. Leave the GAGE. It is acceptable.
B. Calibrate the equipment
C. Calibrate the equipment and check operator variation
D. Calibrate the equipment, check operator variation, and identify Part Variation.

71. The tool that you typically use in the Measure stage to scope a project is:

A. DFMEA
B. PFMEA
C. CE Matrix
D. SIPOC

72. Which one of these is the most important precursor for calculating Capability?

A. Process Stability
B. Normality
C. No skew in data
D. None of the above

73. Which of these measures would you normally associate with Process Capability of Discrete Data?

A. DPMO/PPM
B. DPU
C. Cp
D. Cpk

74. Which of the following methods directly helps you determine batch size?

A. Setup time
B. Changeover time
C. Cycle time
D. EPEI

75. Which type of chart is appropriate when the sample size is variable and each sample may contain more than one instance of the targeted condition?

A. P chart
B. Autocorrelation chart
C. U chart
D. X-bar chart

76. What is the purpose of PERT analysis during the Analyze phase of DMAIC?

A. To identify the most influential steps in a process
B. To monitor improvements in cycle time
C. To identify the critical path of a process
D. To reduce cycle time

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77. Which form of leveling typically includes the use of heijunka boxes?

A. Leveling by volume
B. Leveling by product
C. Demand leveling
D. Customer leveling

78. While performing monthly reviews, in the control chart of one of your production unit, you observe that the last 50 points are very near to the centerline. All of them appear to be within about one sigma of the centerline. What Interpretation would you derive from this?

A. Somebody miscalculated the original calculations of the control limits
B. The process standard deviation has decreased during last month and hence the last 50 data points are showing values much closer to the centerline, But, nobody thought to recompute the control limits to reflect the new reality
C. This is an out of control situation. Get a team in place right away and see what the trouble is.
D. The process is in control. The closer the points are to the centerline the better control of the process is. No action is needed, process and the control limits to continue as it is

79. In an analysis of variance, how is the F statistic used?

A. To compare the mean square treatment with the mean square error
B. To estimate the process average
C. To find the variation within each subgroup
D. To find the variation between different subgroups

80. The flow of activities that include value-added as well as non- value-added activities which take the raw material and deliver the end product to the customer, is called:

A. Value Stream Map
B. Process Analysis
C. SIPOC
D. Non Value Add Activities Analysis

81. Which of the following are Warusa Kagen conditions?

A. Meda, Mera, and Meri
B. Mauda, Maura, and Mauri
C. Muda, Mura, and Muri
D. Moda, Mora, and Mori

82. Tremendous resistance can be expected in which of the Value Stream Analysis Phase(s)?

A. Mura
B. Muda
C. Muri
D. All of the above

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83. What does CLOSEDMITTS stand for?

A. Complexity, Labor, Overproduction, Space, Energy, Defects, Materials, Idle Time, Time, Transportation, Safety Hazards
B. Complexity, Labor, Overproduction, Space, energy, Design, Materials, Innovation, Time, Transportation, Safety Hazards
C. Client, Labor, Overproduction, Stakeholders, Energy, Design, Materials, Innovation, Time, Transportation, Safety Hazards
D. Client, Labor, Optimization, Stakeholders, Energy, Design, Materials, Innovation, Time, Transportation, Safety Measures

84. Spaghetti charts graphically depict the movement of and _____ in a process.

A. Products and people
B. Resources and labor
C. Income and expenses
D. Funds and allocations

85. What does CE diagram stand for?

A. Customer and Engagement
B. Cause and Estimation
C. Client and Expectation
D. Cause and Effect

86. Box Plot is also known as:

A. Whiskers Plot
B. Candlestick Plot
C. Box and Whiskers Plot
D. All of the above

87. Which kind of information does Box plot provide?

A. Location, spread, and variability of data
B. Distance, relativity, and density of data
C. Location, speed, and density of data
D. Continuity, relativity, and variability of data

88. What does KPIV stand for?

A. Key Process Indicator Value
B. Key Person Indicator Value
C. Key Process Input Value
D. Key Person Input Value

89. To design for manufacturability, the following methods are introduced, except:

A. Standardization and simplification
B. Fewer active parts are used
C. Design alternative are used
D. Product quality is held above simplicity

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90. A limitation in Six Sigma implementation is when management applies the methodology:

A. Using trained professionals
B. Using approved funds
C. Only for a certain fixed duration
D. In a few departments within the company.

91. In a scenario where one response variable and one predictor variable are to be tested, which regression analysis needs to be used?

A. Aggregated Linear Regression
B. Single Linear Regression
C. Sample Linear Regression
D. Simple Linear Regression

92. When is Multiple Linear Regression to be used?

A. 1 KPOV and 1 KPIV
B. 1 KPOV and 2 KPIV
C. 2 KPOV and 1 KPIV
D. 2 KPOV and 2 KPIV

93. _____ shows the extent to which multicollinearity exists between predictors.

A. Multiple Linear Regression
B. Variation Inflation Factor
C. Single Linear Regression
D. Statistical Accuracy

94. What tests for the presence of autocorrelation between residuals?

A. Duminy Walker Statistic
B. Durbin Walker Statistic
C. Durbin Watson Statistic
D. Duminy Watson Statistic

95. What does PRESS in PRESS Statistic stand for?

A. Partial Residuals Sum of Squares
B. Part Regression Six Sigma
C. Predicted Residuals Six Sigma
D. Predicted Residuals Sum of Squares

96. Which regression procedures are used in stepwise regression?

A. Standard Stepwise, Forward Selection, and Backward Elimination
B. Standard Stepwise, Forward Estimation, and Backward Estimation
C. Standard Stepwise, Forward Regression, and Backward Regression
D. Standard Stepwise, Forward co-relation, and Backward co-relation

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97. What is Confidence Interval?

A. The interval between 2 consecutive sets of data population
B. Estimated range of values which is likely to include the parameter of the population
C. The interval between each measurement to gain confidence
D. The interval between the Measure and Analyze phase

98. Which factors impact the confidence intervals for sample data?

A. Data Variation and Data Continuity
B. Accuracy and Precision
C. Significance Level and Sample Size
D. Data Relativity

99. Which statistic can be used to determine the probability of variances occurring?

A. Binomial Statistic
B. Poisson Statistic
C. Moving Average Statistic
D. Chi-Square Statistic

100. For a population variance of 1 and sample variance on data collected of 0.6, determine the probability that the sample variance can exceed 0.6. The sample size is 30.

A. A. 15.4
B. B. 16.4
C. C. 17.4
D. D. 18.4

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